

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 7 — Equilibria

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	7.1 Chemical Equilibria · Le Chatelier · K _c & K _p · Industrial Processes · 7.2 Brønsted–Lowry Acids & Bases
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	228 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED**7.1 Chemical equilibria**

— Dynamic equilibrium features, Le Chatelier's principle (concentration, pressure, temperature, catalyst)

7.2 Brønsted–Lowry acids & bases— Proton donors and acceptors, conjugate acid–base pairs, strong vs weak acids/bases, experimental determination of K_a

SUBTOPIC 7.1 — CHEMICAL EQUILIBRIA, LE CHATELIER, K_c & K_p

1 Characteristics of Dynamic Chemical Equilibrium — 9701/22/M/J/21/Q6(a)–(b)

EASY · Syllabus 7.1

Chemical equilibrium is a dynamic state achieved in reversible chemical reactions.

(a) State two essential features of a system that has achieved dynamic chemical equilibrium. [2]

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(b) Explain why dynamic equilibrium can only be maintained indefinitely in a closed system. [1]

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2 Haber Process: Compromise Conditions & Industrial Yield — 9701/21/O/N/20/Q5(a)–(c)

HARD · Syllabus 7.1

Ammonia is synthesized industrially by the Haber process:

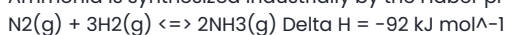


Fig. 7.1 shows the variation of equilibrium percentage yield of ammonia with temperature at different pressures.

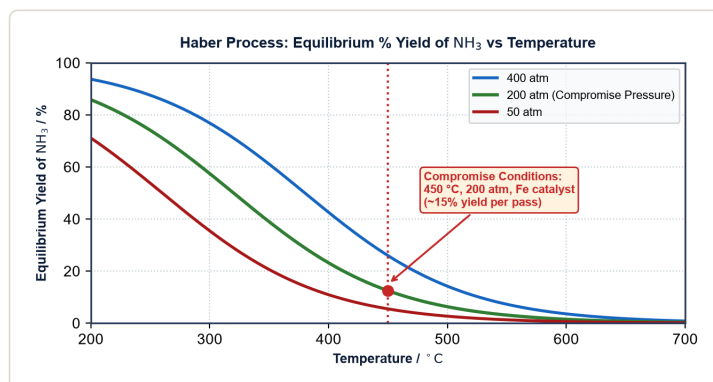


Fig. 7.1: Equilibrium percentage yield of ammonia vs temperature at different operating pressures.

(a) Use Le Chatelier's principle to explain why higher pressures result in a higher equilibrium yield of ammonia. [2]

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(b) Use Le Chatelier's principle to explain why increasing the temperature causes the equilibrium yield of ammonia to decrease. [2]

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(c) In industrial plants, a compromise temperature of 400–450 °C and pressure of 200 atm are employed in the presence of an iron catalyst. Explain why 450 °C is chosen rather than a lower temperature such as 100 °C. [2]

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3 Reaching Dynamic Equilibrium: Rates and Concentrations — 9701/23/M/J/22/Q6(a)–(c)

EASY · Syllabus 7.1

Fig. 7.2 illustrates how the rates of the forward and reverse reactions, and the concentrations of reactants and products, change over time as equilibrium is approached.

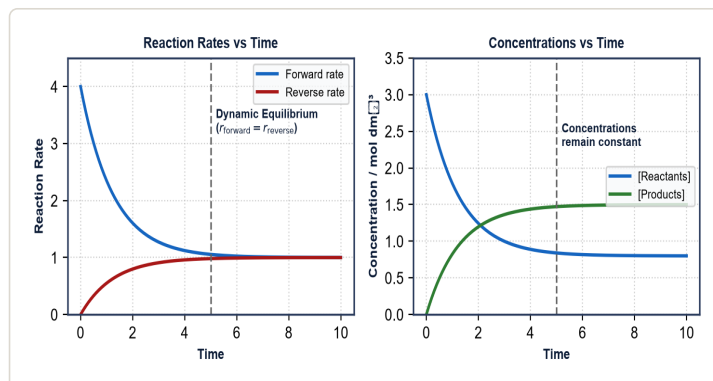


Fig. 7.2: Variation of reaction rates and concentrations over time during approach to dynamic equilibrium.

- (a) Explain why the forward reaction rate is at a maximum at time $t = 0$ and gradually decreases. [2]

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- (b) Explain why the reverse reaction rate starts at zero and gradually increases. [1]

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- (c) State what is true about the forward and reverse reaction rates once dynamic equilibrium has been reached at $t = 5$ minutes. [1]

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4 Le Chatelier Perturbation by Adding Reactant — 9701/22/F/M/21/Q6(a)–(b)

HARD · Syllabus 7.1

Fig. 7.3 shows the effect on an equilibrium mixture of $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ when extra nitrogen gas, N_2 , is injected at $t = 4$ minutes.

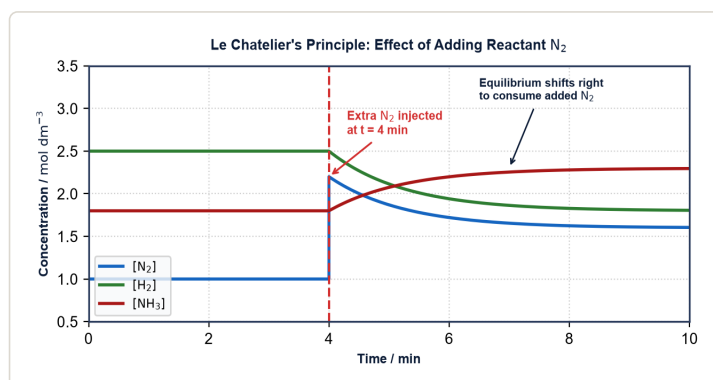


Fig. 7.3: Response of Haber equilibrium concentrations to the injection of extra nitrogen at $t = 4$ min.

- (a) State Le Chatelier's principle. [2]

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- (b) Referring to Fig. 7.3, explain the subsequent changes in the concentrations of H_2 and NH_3 between $t = 4$ minutes and $t = 8$ minutes. [3]

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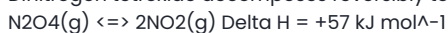
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5 Pressure Effects on Gaseous Equilibrium: $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$ — 9701/21/M/J/22/Q6(a)-(c)

HARD · Syllabus 7.1

Dinitrogen tetroxide decomposes reversibly to nitrogen dioxide in a gas syringe:



(N_2O_4 is colourless; NO_2 is dark brown).

- (a) When the plunger of the syringe is suddenly pushed in, the gas mixture instantaneously turns darker brown, but then gradually fades to a paler brown. Explain both observations. [3]

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- (b) State and explain the effect on the colour of the mixture when the syringe is placed in a beaker of hot water. [2]

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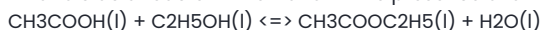
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6 Calculating Concentration Equilibrium Constant, K_c : Esterification — 9701/22/O/N/23/Q6(a)-(c)

HARD · Syllabus 7.1

Ethanoic acid reacts with ethanol in the presence of an acid catalyst to form ethyl ethanoate:



In an experiment, 1.00 mol of CH_3COOH and 1.00 mol of $\text{C}_2\text{H}_5\text{OH}$ were mixed in a sealed tube. At equilibrium at 25°C , 0.667 mol of ethyl ethanoate was present in the total volume V .

- (a) Write the expression for the equilibrium constant K_c for this reaction. [1]

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- (b) Calculate the equilibrium amounts, in moles, of CH_3COOH , $\text{C}_2\text{H}_5\text{OH}$, and H_2O present in the tube. [2]

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- (c) Calculate the value of K_c and state its units, if any. [2]

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7 Calculating Partial Pressure Equilibrium Constant, K_p : PCl_5 Dissociation — 9701/21/M/J/23/Q6(a)-(c)

HARD · Syllabus 7.1

Phosphorus(V) chloride dissociates reversibly at 250°C according to the equation:



A 1.00 mol sample of PCl_5 was heated in a sealed container at 250°C until equilibrium was reached at a total pressure of 1.50×10^5 Pa. At equilibrium, 0.400 mol of Cl_2 was present.

(a) Calculate the total amount, in moles, of all gases present at equilibrium. [2]

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(b) Calculate the partial pressure of each gas in the equilibrium mixture. [2]

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(c) Write the expression for K_p and calculate its value, including units. [2]

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8 Temperature Dependence of Equilibrium Constants — 9701/22/M/J/20/Q6(a)–(b)

EASY · Syllabus 7.1

Consider two reversible reactions with different standard enthalpy changes:

Reaction 1: $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$ $\Delta H = -57 \text{ kJ mol}^{-1}$

Reaction 2: $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$ $\Delta H = +180 \text{ kJ mol}^{-1}$

(a) State and explain the effect of increasing temperature on the numerical value of K_p for Reaction 1. [2]

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(b) State and explain the effect of increasing temperature on the numerical value of K_p for Reaction 2. [2]

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9 Why Pressure and Concentration Changes Do Not Alter K_c or K_p — 9701/21/O/N/22/Q6(a)–(b)

HARD · Syllabus 7.1

Students frequently confuse a shift in equilibrium position with a change in the equilibrium constant.

(a) State the ONLY factor that alters the numerical value of the equilibrium constant K_c for a given reaction. [1]

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(b) Explain why increasing the total pressure of a gaseous system at constant temperature shifts the position of equilibrium without changing the value of K_p . [3]

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10 Contact Process: Industrial Sulfuric Acid Manufacture — 9701/23/O/N/21/Q5(a)–(c)

HARD · Syllabus 7.1

The key stage in the Contact process is the catalytic oxidation of sulfur dioxide:

$2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ $\Delta H = -196 \text{ kJ mol}^{-1}$

Operating conditions: 450°C , 1–2 atm, vanadium(V) oxide catalyst, V_2O_5 .

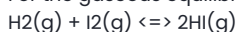
- (a) State the effect of the V_2O_5 catalyst on the position of equilibrium and on the rate of achieving equilibrium. [2]
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- (b) Explain why a relatively low pressure of 1–2 atm is used in the Contact process, even though higher pressure thermodynamically favours SO_3 formation. [2]
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- (c) Explain why SO_3 is absorbed in concentrated H_2SO_4 to form oleum ($\text{H}_2\text{S}_2\text{O}_7$) rather than reacted directly with water. [2]
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11

Calculating Equilibrium Concentrations: $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ — 9701/22/F/M/23/Q6(a)–(c)

HARD · Syllabus 7.1

For the gaseous equilibrium:



The value of K_c is 54.0 at 425 °C.

In an experiment at 425 °C, 1.00 mol of H_2 and 1.00 mol of I_2 were sealed in a 2.00 dm³ vessel.

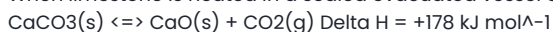
- (a) Let $2x$ be the amount, in moles, of HI formed at equilibrium. Write expressions for the equilibrium concentrations of H_2 , I_2 , and HI in terms of x . [2]
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- (b) Substitute these expressions into K_c and calculate the value of x . [2]
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- (c) Calculate the equilibrium concentration of HI in mol dm⁻³. [1]
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12

Heterogeneous Equilibria: Decomposition of Calcium Carbonate — 9701/21/M/J/21/Q7(a)–(b)

EASY · Syllabus 7.1

When limestone is heated in a sealed evacuated vessel at 800 °C, heterogeneous equilibrium is established:



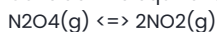
- (a) Write the expression for the equilibrium constant K_p for this heterogeneous reaction, explaining why the solid phases are omitted. [2]
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- (b) At 800 °C, the equilibrium pressure of CO_2 is 24.0 kPa. State the value of K_p with units. [1]
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- (c) State what happens to the equilibrium pressure of CO_2 if the mass of $\text{CaCO}_3(\text{s})$ in the container is doubled at constant temperature. [1]
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13 Effect of Inert Gas Addition on Gaseous Equilibrium — 9701/22/M/J/22/Q6(b)

HARD · Syllabus 7.1

Consider the equilibrium system:



Helium gas is introduced into the container at constant temperature.

- (a) Predict and explain the effect on the position of equilibrium when helium gas is added at constant volume. [2]

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- (b) Predict and explain the effect on the position of equilibrium when helium gas is added at constant total pressure. [2]

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14 Catalysts and Activation Energy in Reversible Reactions — 9701/22/O/N/21/Q7(a)–(b)

EASY · Syllabus 7.1

Catalysts are widely used in chemical manufacture to improve process efficiency.

- (a) Explain why a catalyst has no effect on the equilibrium yield of a reversible reaction. [2]

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- (b) State the commercial benefit of using a catalyst in a reversible reaction. [1]

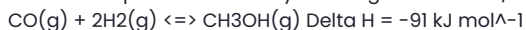
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15 Industrial Synthesis of Methanol: Equilibrium & Economics — 9701/21/O/N/23/Q6(a)–(c)

HARD · Syllabus 7.1

Methanol is produced from synthesis gas over a $\text{Cu/ZnO/Al}_2\text{O}_3$ catalyst:



- (a) State the effect on the equilibrium yield of methanol of increasing pressure and increasing temperature. [2]

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- (b) Write the expression for K_p for this reaction and deduce its units in terms of kPa. [2]

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- (c) In an industrial reactor at 250°C and $5.00 \times 10^3 \text{ kPa}$, the equilibrium partial pressures are: $p(\text{CO}) = 1.20 \times 10^3 \text{ kPa}$, $p(\text{H}_2) = 2.40 \times 10^3 \text{ kPa}$, and $p(\text{CH}_3\text{OH}) = 1.40 \times 10^3 \text{ kPa}$. Calculate K_p . [2]

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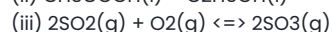
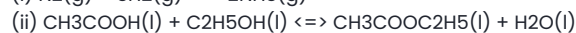
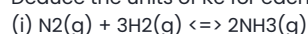
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16 Units of Equilibrium Constants & Dimensional Analysis — 9701/23/M/J/20/Q5(a)–(b)

EASY · Syllabus 7.1

The units of K_c and K_p depend on the stoichiometry of the balanced chemical equation.

(a) Deduce the units of K_c for each of the following equilibria: **[3]**



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(b) Deduce the units of K_p for: $2\text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$. **[1]**

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17 Thermochemistry & Kinetics of NO₂ Dimerisation — 9701/22/F/M/24/Q5(a)–(c)

HARD · Syllabus 7.1

The dimerization of nitrogen dioxide is an exothermic equilibrium:



(a) Draw a labelled reaction coordinate diagram showing the activation energy and ΔH for this reaction. **[2]**

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(b) Explain why the activation energy for the forward reaction ($2\text{NO}_2 \rightarrow \text{N}_2\text{O}_4$) is exceptionally low (near zero). **[1]**

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(c) State how the value of K_c changes when the temperature is raised from 20 °C to 80 °C. **[1]**

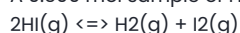
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18 ICE Table Calculation: Dissociation of HI — 9701/21/M/J/24/Q7(a)–(b)

HARD · Syllabus 7.1

A 0.500 mol sample of HI was placed in a 1.00 dm³ flask and allowed to decompose at 700 K:



At equilibrium, 0.105 mol of I₂ was formed.

(a) Construct an ICE table showing the initial, change, and equilibrium moles of all three gases. **[3]**

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(b) Calculate the value of K_c at 700 K. **[2]**

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19 Recycling & Separation in the Haber Process — 9701/22/O/N/24/Q6(a)–(b)

EASY · Syllabus 7.1

In the Haber process, the single-pass conversion of N_2 and H_2 to NH_3 is typically only 15%.

- (a) Explain how ammonia is separated from unreacted nitrogen and hydrogen in the industrial plant. [2]

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- (b) Explain what happens to the unreacted nitrogen and hydrogen gases after separation. [1]

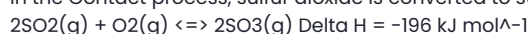
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20 Industrial Equilibrium Control in the Contact Process — 9701/21/O/N/24/Q6(b)

HARD · Syllabus 7.1

In the Contact process, sulfur dioxide is converted to sulfur trioxide in a multi-stage converter:



- (a) Explain why the gas mixture is passed over several successive catalyst beds with intermediate cooling. [2]

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- (b) Explain why an excess of oxygen (air) is used in the feed mixture. [1]

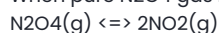
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21 Degree of Dissociation, alpha, in Gaseous Equilibria — 9701/23/M/J/23/Q5(a)–(c)

HARD · Syllabus 7.1

When pure N_2O_4 gas is introduced into an empty chamber at total pressure P , a fraction α dissociates:



- (a) Starting with 1 mole of N_2O_4 , express the equilibrium moles of N_2O_4 , NO_2 , and total moles in terms of α . [2]

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- (b) Show that K_p can be expressed as $K_p = (4 * \alpha^2 * P) / (1 - \alpha^2)$. [2]

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- (c) Calculate the degree of dissociation α at 25 °C if $K_p = 14.0 \text{ kPa}$ and total pressure $P = 100 \text{ kPa}$. [2]

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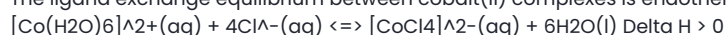
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22 Cobalt Complex Equilibrium: Temperature and Ligand Exchange — 9701/22/M/J/25/Q4(a)–(c)

HARD · Syllabus 7.1

The ligand exchange equilibrium between cobalt(II) complexes is endothermic in the forward direction:



($[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is pink; $[\text{CoCl}_4]^{2-}$ is deep blue).

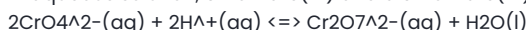
- (a) Predict and explain the colour change when concentrated hydrochloric acid is added to a pink solution of cobalt(II) chloride. [2]
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- (b) Predict and explain the colour change when this mixture is placed into an ice-water bath. [2]
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23 Chromate–Dichromate pH–Dependent Equilibrium — 9701/21/F/M/25/Q6(a)–(b)

EASY · Syllabus 7.1

In aqueous solution, chromate(VI) and dichromate(VI) ions exist in dynamic equilibrium:



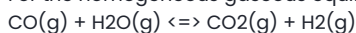
(CrO₄²⁻ is yellow; Cr₂O₇²⁻ is orange).

- (a) State and explain the colour change when dilute sulfuric acid is added to a yellow solution of potassium chromate. [2]
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- (b) State and explain the colour change when excess aqueous sodium hydroxide is subsequently added. [2]
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24 Calculation of K_c When Volume Cancels — 9701/22/O/N/25/Q6(a)–(b)

EASY · Syllabus 7.1

For the homogeneous gaseous equilibrium:



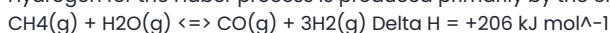
At 800 K, 2.00 mol of CO and 2.00 mol of H₂O were mixed in a vessel of volume V. At equilibrium, 1.33 mol of CO₂ was present.

- (a) Explain why the volume V of the vessel is not needed to calculate the numerical value of K_c. [1]
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- (b) Calculate the equilibrium amounts of all species and determine the value of K_c at 800 K. [3]
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25 Steam Reforming of Methane: Industrial Hydrogen Production — 9701/23/O/N/24/Q5(a)–(c)

HARD · Syllabus 7.1

Hydrogen for the Haber process is produced primarily by the endothermic steam reforming of natural gas:



- (a) Explain how high temperature affects the equilibrium yield of hydrogen. [2]
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- (b) Explain how high pressure affects the equilibrium yield of hydrogen. [2]

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- (c) Suggest why industrial steam reformers operate at moderate pressure (around 30 atm) rather than atmospheric pressure despite the equilibrium prediction. [2]

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26 Dynamic Liquid–Vapour Equilibrium in a Closed Container — 9701/21/M/J/25/Q7(a)–(b)

EASY · Syllabus 7.1

When a sample of liquid water is sealed in an evacuated glass bulb at 25 °C, dynamic equilibrium is established between liquid and vapour.

- (a) Describe dynamic equilibrium at the molecular level in this system. [2]

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- (b) Explain what happens to the vapour pressure inside the bulb when the temperature is raised from 25 °C to 50 °C. [2]

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27 Perturbation of Esterification Equilibrium by Adding Water — 9701/22/M/J/24/Q7(b)

HARD · Syllabus 7.1

Consider the equilibrium:



- (a) State what happens to the yield of ethyl ethanoate if a large excess of water is added to the equilibrium mixture. [1]

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- (b) Explain your answer to (a) using both Le Chatelier's principle and the K_c expression. [3]

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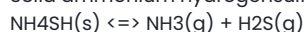
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28 Decomposition of Solid Ammonium Hydrogensulfide — 9701/23/M/J/24/Q6(a)–(b)

HARD · Syllabus 7.1

Solid ammonium hydrogensulfide decomposes endothermically:



When excess $\text{NH}_4\text{SH}(\text{s})$ is placed in an evacuated container at 25 °C, the total equilibrium pressure is 66.0 kPa.

- (a) Deduce the partial pressure of $\text{NH}_3(\text{g})$ and $\text{H}_2\text{S}(\text{g})$ in the container. [2]

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- (b) Calculate the value of K_p at 25 °C, including units. [2]

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29 Reaction Quotient, Q_c , vs Equilibrium Constant, K_c — 9701/22/F/M/25/Q6(b)

HARD · Syllabus 7.1

For the reaction $\text{SO}_2(\text{g}) + \text{NO}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g}) + \text{NO}(\text{g})$, $K_c = 85.0$ at a certain temperature.

A mixture was prepared with concentrations: $[\text{SO}_2] = 0.10$, $[\text{NO}_2] = 0.10$, $[\text{SO}_3] = 0.50$, and $[\text{NO}] = 0.50 \text{ mol dm}^{-3}$.

- (a) Calculate the reaction quotient, Q_c , for this initial mixture. [2]

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- (b) Predict whether the concentration of SO_3 will increase, decrease, or remain unchanged as the system approaches equilibrium. Justify your answer. [2]

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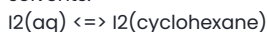
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30 Multi-Phase Distribution Equilibrium (Partition) — 9701/21/O/N/25/Q7(a)-(b)

EASY · Syllabus 7.1

When an aqueous solution of iodine is shaken with cyclohexane in a separating funnel, iodine partitions between the two immiscible solvents:



- (a) State the colours of the aqueous layer and the cyclohexane layer at equilibrium. [2]

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- (b) Explain why iodine dissolves preferentially in non-polar cyclohexane rather than in polar water. [2]

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SUBTOPIC 7.2 — BRØNSTED-LOWRY ACIDS & BASES

31 Brønsted-Lowry Theory & Conjugate Acid-Base Pairs — 9701/22/M/J/21/Q7(a)-(c)

EASY · Syllabus 7.2

Fig. 7.4 illustrates the Brønsted-Lowry concept of proton transfer and conjugate acid-base pairs for ethanoic acid in water.

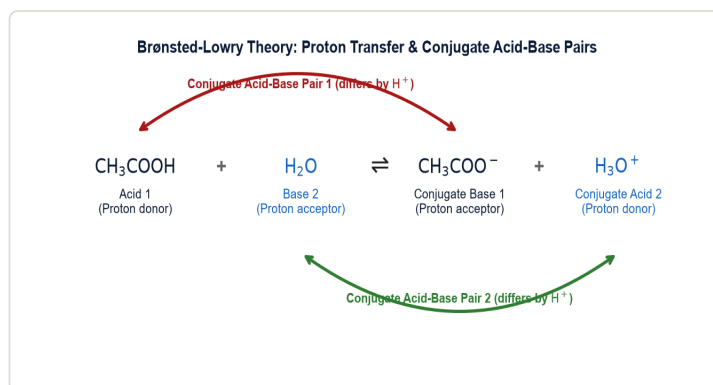


Fig. 7.4: Proton transfer and conjugate acid-base pairs in Brønsted-Lowry theory.

- (a) Define a Brønsted–Lowry acid and a Brønsted–Lowry base. [2]
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- (b) In the reaction shown in Fig. 7.4, identify the conjugate base of CH_3COOH and the conjugate acid of H_2O . [2]
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- (c) Explain why the species in a conjugate acid–base pair always differ by exactly one H^+ ion. [1]
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32 Identifying Conjugate Pairs in Complex Acid–Base Systems — 9701/21/O/N/21/Q6(a)–(b)

EASY · Syllabus 7.2

Consider the two acid–base equilibria:

Equilibrium 1: $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$

Equilibrium 2: $\text{HNO}_3 + \text{H}_2\text{SO}_4 \rightleftharpoons \text{H}_2\text{NO}_3^+ + \text{HSO}_4^-$

- (a) For Equilibrium 1, identify the two conjugate acid–base pairs. [2]
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- (b) For Equilibrium 2, identify which species acts as the Brønsted–Lowry base and explain why. [2]
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33 Strong vs Weak Acids: Degree of Dissociation & pH — 9701/23/M/J/22/Q7(a)–(c)

HARD · Syllabus 7.2

Hydrochloric acid and ethanoic acid are monoprotic acids.

- (a) Define the difference between a strong acid and a weak acid in terms of dissociation. [2]
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- (b) Calculate the pH of $0.0500 \text{ mol dm}^{-3} \text{ HCl}(\text{aq})$ given that $\text{pH} = -\log_{10}[\text{H}^+]$. [2]
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- (c) State and explain whether $0.0500 \text{ mol dm}^{-3} \text{ CH}_3\text{COOH}(\text{aq})$ will have a higher, lower, or identical pH compared to $0.0500 \text{ mol dm}^{-3} \text{ HCl}(\text{aq})$. [2]
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34 Experimental Comparison of Strong and Weak Acids — 9701/22/F/M/22/Q5(a)–(c)

EASY · Syllabus 7.2

A student is provided with two unlabelled bottles, one containing 0.10 mol dm^{-3} HCl and the other containing 0.10 mol dm^{-3} CH₃COOH.

- (a) Describe how measuring the electrical conductivity can distinguish between the two acids. [2]

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- (b) Describe what is observed when identical strips of magnesium ribbon are added to equal volumes of each acid. [2]

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- (c) State whether equal volumes of both acids would require the same volume or different volumes of 0.10 mol dm^{-3} NaOH for complete neutralisation. Justify your answer. [2]

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35 Amphiprotic Behaviour of Water and Hydrogencarbonate — 9701/21/M/J/22/Q7(a)–(b)

EASY · Syllabus 7.2

Certain chemical species can act as either a Brønsted–Lowry acid or a Brønsted–Lowry base depending on the reaction partner.

- (a) Write two chemical equations demonstrating that water, H₂O, is amphiprotic (amphoteric). [2]
- (i) Acting as an acid with NH₃
- (ii) Acting as a base with HCl

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- (b) Write two equations showing the amphiprotic behaviour of the hydrogencarbonate ion, HCO₃[–]. [2]

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36 Strong vs Weak Bases: Sodium Hydroxide vs Ammonia — 9701/22/O/N/22/Q5(a)–(b)

EASY · Syllabus 7.2

Aqueous sodium hydroxide, NaOH, and aqueous ammonia, NH₃, are both basic solutions.

- (a) Write equations showing how NaOH and NH₃ generate hydroxide ions in aqueous solution. [2]

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- (b) Explain why 0.10 mol dm^{-3} NaOH has a pH of approximately 13 while 0.10 mol dm^{-3} NH₃ has a pH of approximately 11. [2]

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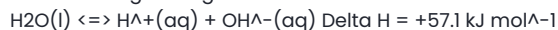
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37 Ionic Product of Water, Kw & Neutrality — 9701/23/O/N/22/Q5(a)–(c)

HARD · Syllabus 7.2

Water undergoes slight autoionisation:



At 25 °C, $K_w = [\text{H}^+][\text{OH}^-] = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$.

- (a)** State the mathematical definition of neutral pH in pure water. **[1]**

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- (b)** Explain why the value of Kw increases as the temperature of water is raised from 25 °C to 60 °C. **[2]**

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- (c)** At 60 °C, $K_w = 9.55 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Calculate the pH of pure water at 60 °C and state whether pure water at 60 °C is acidic, alkaline, or neutral. **[3]**

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38 pH Calculations for Strong Monoprotic and Diprotic Acids — 9701/22/F/M/23/Q7(a)–(b)

EASY · Syllabus 7.2

For strong acids, complete dissociation is assumed in aqueous solution.

- (a)** Calculate the pH of 0.0250 mol dm⁻³ nitric acid, HNO₃. **[2]**

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- (b)** Assuming complete dissociation of both protons, calculate the pH of 0.0250 mol dm⁻³ sulfuric acid, H₂SO₄. **[2]**

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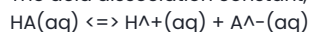
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39 Relative Acid Strengths of Carboxylic Acids — 9701/21/M/J/23/Q7(a)–(b)

HARD · Syllabus 7.2

The acid dissociation constant, K_a, measures the extent of dissociation of a weak acid:



• Methanoic acid, HCOOH: $K_a = 1.8 \times 10^{-4} \text{ mol dm}^{-3}$

• Ethanoic acid, CH₃COOH: $K_a = 1.7 \times 10^{-5} \text{ mol dm}^{-3}$

- (a)** State which of these two acids is stronger and explain your answer by referring to the K_a values. **[2]**

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- (b)** Explain why ethanoic acid is a weaker acid than methanoic acid in terms of the electron-donating inductive effect of the methyl group. **[2]**

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40 Zwitterions & Amphoteric Nature of Amino Acids — 9701/22/M/J/23/Q5(a)–(c)

HARD · Syllabus 7.2

Glycine, $\text{H}_2\text{N}-\text{CH}_2-\text{COOH}$, is the simplest amino acid. In aqueous solution, it exists predominantly as a zwitterion: $^+\text{H}_3\text{N}-\text{CH}_2-\text{COO}^-$.

(a) Define the term zwitterion. **[1]**

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(b) Write the structural formula of the organic species formed when glycine zwitterion reacts with dilute hydrochloric acid. **[1]**

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(c) Write the structural formula of the organic species formed when glycine zwitterion reacts with aqueous sodium hydroxide. **[1]**

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41 Neutralisation Titration Curves: Strong vs Weak Acid — 9701/21/O/N/23/Q7(a)–(c)

HARD · Syllabus 7.2

Consider the titration of 25.0 cm^3 of 0.10 mol dm^{-3} acid with 0.10 mol dm^{-3} NaOH:

- Curve A: 0.10 mol dm^{-3} HCl
- Curve B: 0.10 mol dm^{-3} CH_3COOH

(a) State the initial pH for Curve A and Curve B. **[2]**

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(b) State the pH at the equivalence point for Curve A and for Curve B. **[2]**

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(c) Explain why the equivalence point pH for Curve B is greater than 7.0. **[2]**

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42 Selection of Appropriate Acid-Base Indicators — 9701/22/O/N/23/Q7(b)

EASY · Syllabus 7.2

Indicators change colour over a characteristic pH transition range:

- Methyl orange: pH 3.1 to 4.4 (red to yellow)
- Phenolphthalein: pH 8.3 to 10.0 (colourless to pink)

(a) State which indicator, methyl orange or phenolphthalein, is suitable for titrating ethanoic acid with sodium hydroxide. Justify your choice. **[2]**

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(b) State which indicator is suitable for titrating aqueous ammonia with hydrochloric acid. Justify your choice. **[2]**

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43 Qualitative Action of Buffer Solutions — 9701/23/M/J/24/Q7(a)–(b)

EASY · Syllabus 7.2

A buffer solution resists changes in pH when small amounts of acid or base are added.

- (a) State the two components of an acidic buffer solution. [2]

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- (b) Explain, with the aid of an ionic equation, how an ethanoic acid / sodium ethanoate buffer removes added H^+ ions. [2]

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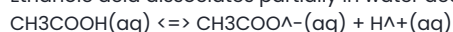
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44 Expression and Units of Acid Dissociation Constant, K_a — 9701/21/M/J/24/Q8(a)–(b)

EASY · Syllabus 7.2

Ethanoic acid dissociates partially in water according to the equilibrium:



- (a) Write the expression for the acid dissociation constant, K_a , for ethanoic acid and deduce its units. [2]

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- (b) Explain why the concentration of water is not included in the K_a expression. [1]

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45 Acid–Base Character of Period 3 Oxides — 9701/22/M/J/24/Q8(a)–(c)

HARD · Syllabus 7.2

The nature of oxides across Period 3 changes from basic to amphoteric to acidic:



- (a) Classify the acid–base nature of Na_2O , Al_2O_3 , and SO_3 . [3]

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- (b) Write a chemical equation showing the reaction of basic Na_2O with water, and state the approximate pH of the resulting solution. [2]

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- (c) Write two equations demonstrating the amphoteric nature of Al_2O_3 reacting with an acid and with an alkali. [2]

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46 Enthalpy of Neutralisation & Weak Acid Dissociation Energetics — 9701/23/O/N/23/Q6(a)–(b)

HARD · Syllabus 7.2

Consider the standard enthalpies of neutralisation:

- Reaction 1: $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$ $\Delta H_{\text{neut}} = -57.1 \text{ kJ mol}^{-1}$
- Reaction 2: $\text{HCN}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCN}(\text{aq}) + \text{H}_2\text{O}(\text{l})$ $\Delta H_{\text{neut}} = -11.7 \text{ kJ mol}^{-1}$

- (a) Explain why Reaction 2 is significantly less exothermic than Reaction 1. [2]

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- (b) Calculate the enthalpy change of dissociation of hydrocyanic acid: $\text{HCN(aq)} \rightarrow \text{H}^+(\text{aq}) + \text{CN}^-(\text{aq})$. [2]

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47 Inductive Effects on Carboxylic Acid Strength — 9701/22/F/M/25/Q7(a)-(b)

HARD · Syllabus 7.2

The acid strength of substituted ethanoic acids increases markedly with halogen substitution:

- Ethanoic acid, CH_3COOH : $K_a = 1.7 \times 10^{-5} \text{ mol dm}^{-3}$
- Chloroethanoic acid, CH_2ClCOOH : $K_a = 1.3 \times 10^{-3} \text{ mol dm}^{-3}$
- Trichloroethanoic acid, CCl_3COOH : $K_a = 2.2 \times 10^{-1} \text{ mol dm}^{-3}$

- (a) Explain why chloroethanoic acid is a significantly stronger acid than ethanoic acid. [3]

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- (b) Predict and explain whether fluoroethanoic acid, CH_2FCOOH , is stronger or weaker than chloroethanoic acid. [2]

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48 Salt Hydrolysis & Solution pH — 9701/21/F/M/25/Q7(a)-(c)

HARD · Syllabus 7.2

When salts dissolve in water, the resulting solutions are not always neutral.

- (a) Predict whether an aqueous solution of ammonium chloride, NH_4Cl , will have a $\text{pH} < 7$, $\text{pH} = 7$, or $\text{pH} > 7$. Explain your answer with an equation. [2]

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- (b) Predict whether an aqueous solution of sodium ethanoate, CH_3COONa , will have a $\text{pH} < 7$, $\text{pH} = 7$, or $\text{pH} > 7$. Explain your answer with an equation. [2]

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49 Base Strength of Nitrogen Compounds — 9701/22/M/J/25/Q7(a)-(b)

HARD · Syllabus 7.2

The base strength of nitrogen bases depends on the availability of the lone pair of electrons on the nitrogen atom:
Ammonia, NH_3 ; Methylamine, CH_3NH_2 ; Phenylamine, $\text{C}_6\text{H}_5\text{NH}_2$.

- (a) Explain why methylamine is a stronger base than ammonia. [2]

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- (b) Explain why phenylamine is a much weaker base than ammonia. [2]

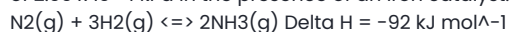
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SUBTOPIC 7.1 — CHEMICAL EQUILIBRIA, LE CHATELIER, K_c & K_p **50 Comprehensive Synoptic: Haber Process Equilibria, K_p & Basic Character of Ammonia — 9701/22/O/N/25/Q7(a)-(d) [4] Syllabus 7.1**

In an industrial synthesis, an initial equimolar mixture of N_2 and H_2 was allowed to reach equilibrium at $450\text{ }^\circ\text{C}$ and a total pressure of $2.00 \times 10^4\text{ kPa}$ in the presence of an iron catalyst:



At equilibrium, the mole fraction of ammonia is 0.150.

- (a) Calculate the partial pressure of ammonia, $p(NH_3)$, at equilibrium. [1]

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- (b) Given that the remaining gas mixture maintains the initial 1 N_2 : 3 H_2 molar ratio, calculate the partial pressures of N_2 and H_2 , and calculate K_p . [3]

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- (c) Ammonia dissolves in water to form an alkaline solution. Write the equation for this reaction, identifying the conjugate acid and conjugate base. [2]

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- (d) Explain how Le Chatelier's principle and reaction kinetics dictate the compromise operating conditions chosen for the industrial manufacture of ammonia. [2]

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Mark Scheme — Topic 7 — Equilibria

CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	The rate of the forward reaction equals the rate of the reverse reaction (1); The concentrations of reactants and products remain constant over time (or macroscopic properties like color/pressure remain constant) (1)	2
1(b)	In an open system, matter / gaseous reactants or products can escape to the surroundings, preventing equilibrium from being established (1)	1
2(a)	There are 4 moles of gas on the reactant side and 2 moles of gas on the product side (1); An increase in pressure shifts equilibrium to the right towards fewer moles of gas to relieve the pressure (1)	2
2(b)	The forward reaction is exothermic ($\Delta H = -92 \text{ kJ mol}^{-1}$) (1); Increasing temperature shifts equilibrium in the endothermic reverse direction to absorb the added heat (1)	2
2(c)	At 100 °C, the rate of reaction is far too slow to be commercially viable because few particles possess energy $> E_a$ (1); 450 °C provides an acceptable reaction rate while still achieving a viable equilibrium yield (~15% per pass) (1)	2
3(a)	At $t = 0$, reactant concentrations are at their highest, leading to maximum collision frequency (1); As reactants are consumed, their concentrations decrease, reducing collision frequency and rate (1)	2
3(b)	At $t = 0$, no product molecules exist; as products are formed, their concentration increases, allowing reverse collisions to occur at an increasing rate (1)	1
3(c)	The forward rate and reverse rate become exactly equal ($r_{\text{forward}} = r_{\text{reverse}}$) (1)	1
4(a)	If a system at equilibrium is subjected to a change in conditions (concentration, temperature, or pressure) (1); the position of equilibrium shifts in a direction that tends to oppose / counteract the effect of that change (1)	2
4(b)	The sudden increase in $[N_2]$ causes equilibrium to shift to the right to consume the added N_2 (1); As the forward reaction proceeds, $[H_2]$ decreases as it reacts with N_2 (1); Simultaneously, $[NH_3]$ increases until a new dynamic equilibrium is established (1)	3
5(a)	Instantly compressing the gas decreases volume, so concentration of all species (including brown NO_2) increases immediately, darkening the color (1); Higher pressure shifts equilibrium to the left towards fewer moles of gas (1 N_2O_4 : 2 NO_2) to reduce pressure (1); Some brown NO_2 converts into colourless N_2O_4 , causing the colour to lighten gradually (1)	3
5(b)	The mixture turns darker brown (1); The forward reaction is endothermic ($\Delta H = +57 \text{ kJ mol}^{-1}$), so increasing temperature shifts equilibrium to the right to absorb heat, increasing $[NO_2]$ (1)	2
6(a)	$K_c = \frac{[CH_3COOC_2H_5][H_2O]}{[CH_3COOH][C_2H_5OH]}$ (1)	1
6(b)	$n(CH_3COOC_2H_5) = 0.667 \text{ mol}$, $n(H_2O) = 0.667 \text{ mol}$ (1); $n(CH_3COOH) = 1.00 - 0.667 = 0.333 \text{ mol}$, $n(C_2H_5OH) = 1.00 - 0.667 = 0.333 \text{ mol}$ (1)	2
6(c)	$K_c = \frac{(0.667/V \times 0.667/V)}{(0.333/V \times 0.333/V)} = \frac{(0.667 \times 0.667)}{(0.333 \times 0.333)} = 4.01$ (allow 4.0) (1); Units: No units / dimensionless (V cancels out completely) (1)	2
7(a)	At equilibrium: $n(Cl_2) = 0.400 \text{ mol}$, $n(PCI_3) = 0.400 \text{ mol}$, $n(PCI_5) = 1.00 - 0.400 = 0.600 \text{ mol}$ (1); Total moles $n_{\text{tot}} = 0.600 + 0.400 + 0.400 = 1.400 \text{ mol}$ (1)	2
7(b)	$p(PCI_5) = \frac{(0.600/1.400) \times 1.50 \times 10^5}{1} = 6.43 \times 10^4 \text{ Pa}$; $p(PCI_3) = p(Cl_2) = \frac{(0.400/1.400) \times 1.50 \times 10^5}{1} = 4.29 \times 10^4 \text{ Pa}$ (2)	2
7(c)	$K_p = \frac{(p(PCI_3) \times p(Cl_2))}{p(PCI_5)} = \frac{(4.29 \times 10^4)^2}{(6.43 \times 10^4)} = 2.86 \times 10^4 \text{ Pa}$ (allow 2.85 – 2.87 $\times 10^4 \text{ Pa}$ / 28.6 kPa) (1); Units: Pa (or kPa) (1)	2
8(a)	K_p decreases (1); Reaction 1 is exothermic, so increasing temperature shifts equilibrium to the left, decreasing product partial pressure and increasing reactant partial pressure (1)	2
8(b)	K_p increases (1); Reaction 2 is endothermic, so increasing temperature shifts equilibrium to the right, increasing product partial pressure relative to reactants (1)	2
9(a)	Temperature (1)	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
9(b)	Increasing total pressure initially causes the partial pressures of all gases to increase proportionally (1); The partial pressure quotient Q_p momentarily deviates from K_p (1); The partial pressures of individual gases adjust as the reaction shifts towards the side with fewer gas molecules until the ratio exactly equals the original K_p value again (1)	3
10(a)	No effect on the position of equilibrium / yield of SO_3 (1); Increases the rate of reaching equilibrium by lowering activation energy of both forward and reverse reactions equally (1)	2
10(b)	The equilibrium yield of SO_3 is already extremely high (~98-99%) at 1-2 atm (1); Operating at higher pressures would require expensive thick-walled pipes, compressors, and maintenance with negligible increase in yield (1)	2
10(c)	Direct reaction of SO_3 with water is violently exothermic and creates a hazardous, uncontrollable mist/fog of sulfuric acid droplets (1); Dissolving SO_3 in concentrated H_2SO_4 to form oleum is safely controllable (1)	2
11(a)	$[\text{H}_2] = (1.00 - x) / 2.00 \text{ mol dm}^{-3}$; $[\text{I}_2] = (1.00 - x) / 2.00 \text{ mol dm}^{-3}$; $[\text{HI}] = 2x / 2.00 = x \text{ mol dm}^{-3}$ (2)	2
11(b)	$K_c = (2x)^2 / (1.00 - x)^2 = 54.0 \Rightarrow 2x / (1.00 - x) = \sqrt{54.0} = 7.348$ (1); $2x = 7.348 - 7.348x \Rightarrow 9.348x = 7.348 \Rightarrow x = 0.786 \text{ mol (1)}$	2
11(c)	$[\text{HI}] = 2x / 2.00 = x = 0.786 \text{ mol dm}^{-3}$ (1)	1
12(a)	$K_p = p(\text{CO}_2)$ (1); Pure solids have constant concentrations / activities of 1 that do not vary with amount, so they are incorporated into the constant (1)	2
12(b)	$K_p = 24.0 \text{ kPa}$ (or $2.40 \times 10^4 \text{ Pa}$) (1)	1
12(c)	No change in CO_2 pressure (K_p depends only on temperature) (1)	1
13(a)	No effect on equilibrium position (1); Adding helium at constant volume increases total pressure but does not change the partial pressures or concentrations of N_2O_4 or NO_2 (1)	2
13(b)	Equilibrium shifts to the right (towards 2NO_2) (1); To maintain constant total pressure, container volume must expand, reducing the partial pressures of N_2O_4 and NO_2 ; equilibrium shifts towards more moles of gas (1)	2
14(a)	A catalyst lowers the activation energy of the forward and reverse reactions by the exact same amount (1); Therefore, the rates of both forward and reverse reactions are accelerated by the same factor (1)	2
14(b)	Equilibrium is reached in a much shorter time / allows the process to run at lower temperatures, reducing fuel and operating costs (1)	1
15(a)	Increasing pressure increases yield (3 moles gas $\rightarrow$ 1 mole gas) (1); Increasing temperature decreases yield (forward reaction is exothermic) (1)	2
15(b)	$K_p = p(\text{CH}_3\text{OH}) / [p(\text{CO}) \times (p(\text{H}_2))^2]$ (1); Units: kPa^{-2} (or Pa^{-2}) (1)	2
15(c)	$K_p = 1.40 \times 10^3 / [1.20 \times 10^3 \times (2.40 \times 10^3)^2] = 1.40 \times 10^3 / [1.20 \times 10^3 \times 5.76 \times 10^6]$ (1); $K_p = 2.03 \times 10^{-7} \text{ kPa}^{-2}$ (allow $2.02 - 2.04 \times 10^{-7} \text{ kPa}^{-2}$) (1)	2
16(a)	(i) $\text{mol}^{-2} \text{ dm}^6$ (1); (ii) No units / dimensionless (1); (iii) $\text{mol}^{-1} \text{ dm}^3$ (1)	3
16(b)	No units / dimensionless ($\text{Pa}^2 / \text{Pa}^2$ cancels out) (1)	1
17(a)	Reactants 2NO_2 at higher enthalpy than product N_2O_4 with downward arrow labelled $\Delta H = -57.2 \text{ kJ mol}^{-1}$ (1); Very low energy barrier from reactants to transition state (1)	2
17(b)	NO_2 is a free radical with an unpaired electron; bond formation between two unpaired electrons requires no bond breaking (1)	1
17(c)	K_c decreases (forward reaction is exothermic) (1)	1
18(a)	Initial: $\text{HI} = 0.500$, $\text{H}_2 = 0$, $\text{I}_2 = 0$ (1); Change: $\text{HI} = -2(0.105) = -0.210$, $\text{H}_2 = +0.105$, $\text{I}_2 = +0.105$ (1); Equilibrium: $\text{HI} = 0.290$, $\text{H}_2 = 0.105$, $\text{I}_2 = 0.105 \text{ mol (1)}$	3
18(b)	$K_c = ([\text{H}_2][\text{I}_2]) / [\text{HI}]^2 = (0.105 \times 0.105) / (0.290)^2$ (1); $K_c = 0.011025 / 0.0841 = 0.0131$ (allow $0.0131 - 0.0132$) (1)	2
19(a)	The gases are cooled under pressure (1); Ammonia has a much higher boiling point (-33°C) due to hydrogen bonding and readily liquefies, while N_2 and H_2 remain gaseous and are drawn off (1)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
19(b)	Unreacted N ₂ and H ₂ gases are recycled back into the reactor, achieving an overall conversion rate of ~98% (1)	1
20(a)	As the exothermic reaction proceeds, temperature rises, which unfavourably shifts equilibrium backwards and limits yield (1); Intermediate cooling removes heat, shifting equilibrium forward in the next bed to maximise overall conversion (1)	2
20(b)	Excess O ₂ drives the equilibrium to the right (Le Chatelier's principle) to maximise conversion of expensive SO ₂ (1)	1
21(a)	$n(\text{N}_2\text{O}_4) = 1 - \alpha$, $n(\text{NO}_2) = 2\alpha$ (1); Total moles $n_{\text{tot}} = (1 - \alpha) + 2\alpha = 1 + \alpha$ (1)	2
21(b)	$p(\text{NO}_2) = [2\alpha / (1 + \alpha)] P$, $p(\text{N}_2\text{O}_4) = [(1 - \alpha) / (1 + \alpha)] P$ (1); $K_p = p(\text{NO}_2)^2 / p(\text{N}_2\text{O}_4) = 4\alpha^2 P^2 / [(1 + \alpha)^2] / [(1 - \alpha)P / (1 + \alpha)] = 4\alpha^2 P / (1 - \alpha^2)$ (1)	2
21(c)	$14.0 = (4 \times \alpha^2 \times 100) / (1 - \alpha^2) \Rightarrow 14(1 - \alpha^2) = 400\alpha^2 \Rightarrow 14 = 414\alpha^2$ (1); $\alpha^2 = 14 / 414 = 0.0338 \Rightarrow \alpha = 0.184$ (or 18.4% dissociation) (1)	2
22(a)	Turns deep blue (1); Adding conc HCl increases [Cl ⁻], shifting equilibrium to the right to consume Cl ⁻ ions, forming blue [CoCl ₄] ²⁻ (1)	2
22(b)	Turns pink (1); Forward reaction is endothermic ($\Delta H > 0$), so cooling shifts equilibrium in the exothermic reverse direction to release heat, reforming pink [Co(H ₂ O) ₆] ²⁺ (1)	2
23(a)	Turns orange (1); Adding acid increases [H ⁺], which shifts equilibrium to the right to consume H ⁺ , forming orange Cr ₂ O ₇ ²⁻ (1)	2
23(b)	Turns yellow (1); OH ⁻ ions react with H ⁺ to form H ₂ O (neutralisation), reducing [H ⁺] and shifting equilibrium to the left to replace H ⁺ , reforming yellow CrO ₄ ²⁻ (1)	2
24(a)	The number of moles of gaseous products (1 + 1 = 2) equals the number of moles of gaseous reactants (1 + 1 = 2), so volume terms cancel out completely in K _c (1)	1
24(b)	At equilibrium: $n(\text{CO}_2) = 1.33$ mol, $n(\text{H}_2) = 1.33$ mol; $n(\text{CO}) = 2.00 - 1.33 = 0.67$ mol, $n(\text{H}_2\text{O}) = 2.00 - 1.33 = 0.67$ mol (1); $K_c = (1.33 \times 1.33) / (0.67 \times 0.67)$ (1); $K_c = 3.94$ (allow 3.9 – 4.0) (1)	3
25(a)	Increases hydrogen yield (1); Forward reaction is endothermic ($\Delta H = +206$ kJ mol ⁻¹), so high temperature shifts equilibrium right to absorb heat (1)	2
25(b)	Decreases hydrogen yield (1); There are 2 moles of gas on reactant side and 4 moles on product side, so high pressure shifts equilibrium left towards fewer moles of gas (1)	2
25(c)	Higher pressure greatly increases reaction rate (more frequent collisions) (1); Keeps gas volume manageable and avoids costly re-compression of hydrogen for subsequent high-pressure ammonia synthesis (1)	2
26(a)	Water molecules continuously evaporate from liquid surface at the same rate that vapour molecules condense back into liquid (1); The mass of liquid and mass of vapour remain constant over time (1)	2
26(b)	Vapour pressure increases significantly (1); Higher temperature gives more molecules kinetic energy exceeding the intermolecular forces, increasing evaporation rate until a higher equilibrium vapour pressure is reached (1)	2
27(a)	Yield of ethyl ethanoate decreases / ester hydrolyses (1)	1
27(b)	By Le Chatelier's principle, adding water increases [H ₂ O], shifting equilibrium to the left to consume water (1); In terms of K _c , adding water makes the reaction quotient $Q_c = ([\text{ester}][\text{water}] / ([\text{acid}][\text{alcohol}])$ greater than K _c (1); Net reverse reaction occurs until Q _c decreases back to equal K _c = 4.0 (1)	3
28(a)	Gases are produced in 1:1 equimolar ratio, so $p(\text{NH}_3) = p(\text{H}_2\text{S}) = 66.0 / 2 = 33.0$ kPa (2)	2
28(b)	$K_p = p(\text{NH}_3) \times p(\text{H}_2\text{S}) = 33.0 \times 33.0 = 1089$ kPa ² (or 1.09×10^3 kPa ² / 1.09×10^9 Pa ²) (1); Units: kPa ² (or Pa ²) (1)	2
29(a)	$Q_c = ([\text{SO}_3][\text{NO}]) / ([\text{SO}_2][\text{NO}_2]) = (0.50 \times 0.50) / (0.10 \times 0.10)$ (1); $Q_c = 0.25 / 0.01 = 25.0$ (1)	2
29(b)	Concentration of SO ₃ will increase (1); Since Q _c (25.0) < K _c (85.0), the system must shift to the right to form more products until Q _c reaches 85.0 (1)	2
30(a)	Aqueous layer: Pale yellow / brown (1); Cyclohexane layer: Purple / violet (1)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
30(b)	Iodine is a non-polar molecule that forms London dispersion forces with non-polar cyclohexane molecules (1); In water, dissolving iodine requires disrupting strong hydrogen bonds between water molecules without forming strong solute-solvent attractions (1)	2
31(a)	Brønsted-Lowry acid: A proton (H ⁺) donor (1); Brønsted-Lowry base: A proton (H ⁺) acceptor (1)	2
31(b)	Conjugate base of CH ₃ COOH: Ethanoate ion, CH ₃ COO ⁻ (1); Conjugate acid of H ₂ O: Hydronium ion, H ₃ O ⁺ (1)	2
31(c)	When an acid donates one proton it becomes its conjugate base, and when a base accepts one proton it becomes its conjugate acid (1)	1
32(a)	Pair 1: H ₂ O (acid) and OH ⁻ (conjugate base) (1); Pair 2: NH ₄ ⁺ (conjugate acid) and NH ₃ (base) (1)	2
32(b)	HNO ₃ acts as the Brønsted-Lowry base (1); It accepts a proton (H ⁺) from the stronger acid H ₂ SO ₄ to form H ₂ NO ₃ ⁺ (1)	2
33(a)	A strong acid is completely (100%) dissociated into ions in aqueous solution (1); A weak acid is only partially dissociated into ions in aqueous solution (1)	2
33(b)	For strong acid, [H ⁺] = 0.0500 mol dm ⁻³ ; pH = -log ₁₀ (0.0500) (1); pH = 1.30 (1)	2
33(c)	Higher pH (1); Ethanoic acid is only partially dissociated, so [H ⁺] is much lower than 0.0500 mol dm ⁻³ , resulting in a less acidic solution with a higher pH value (approx 3.0) (1)	2
34(a)	HCl has significantly higher electrical conductivity than CH ₃ COOH (1); HCl is fully dissociated and contains a much higher concentration of mobile charge-carrying ions (H ⁺ and Cl ⁻) (1)	2
34(b)	Both produce effervescence/bubbles of H ₂ gas (1); Reaction is much faster / more vigorous with HCl due to much higher initial [H ⁺] (1)	2
34(c)	Exactly the same volume of NaOH is required (1); Both solutions contain the same total amount (moles) of neutralisable acidic protons; as OH ⁻ reacts, weak acid equilibrium shifts completely right until fully neutralised (1)	2
35(a)	(i) H ₂ O + NH ₃ ⇌ OH ⁻ + NH ₄ ⁺ (1); (ii) H ₂ O + HCl → H ₃ O ⁺ + Cl ⁻ (1)	2
35(b)	Acid: HCO ₃ ⁻ + OH ⁻ → CO ₃ ²⁻ + H ₂ O (1); Base: HCO ₃ ⁻ + H ⁺ → H ₂ O + CO ₂ (or H ₂ CO ₃) (1)	2
36(a)	NaOH(s) + aq → Na ⁺ (aq) + OH ⁻ (aq) (1); NH ₃ (aq) + H ₂ O(l) ⇌ NH ₄ ⁺ (aq) + OH ⁻ (aq) (1)	2
36(b)	NaOH is a strong base that fully dissociates to give [OH ⁻] = 0.10 mol dm ⁻³ (1); NH ₃ is a weak base that only partially reacts with water, giving a much lower equilibrium [OH ⁻] (~10 ⁻³ mol dm ⁻³) and hence lower pH (1)	2
37(a)	[H ⁺] = [OH ⁻] (1)	1
37(b)	Autoionisation of water is an endothermic process (ΔH > 0) (1); Increasing temperature shifts equilibrium to the right to absorb heat, increasing [H ⁺] and [OH ⁻] and hence K _w (1)	2
37(c)	[H ⁺] = sqrt(K _w) = sqrt(9.55 × 10 ⁻¹⁴) = 3.09 × 10 ⁻⁷ mol dm ⁻³ (1); pH = -log ₁₀ (3.09 × 10 ⁻⁷) = 6.51 (1); It is NEUTRAL because [H ⁺] is still strictly equal to [OH ⁻] (1)	3
38(a)	[H ⁺] = 0.0250 mol dm ⁻³ ; pH = -log ₁₀ (0.0250) (1); pH = 1.60 (1)	2
38(b)	H ₂ SO ₄ → 2H ⁺ + SO ₄ ²⁻ ⇒ [H ⁺] = 2 × 0.0250 = 0.0500 mol dm ⁻³ (1); pH = -log ₁₀ (0.0500) = 1.30 (1)	2
39(a)	Methanoic acid is stronger (1); It has a larger K _a value (1.8 × 10 ⁻⁴ vs 1.7 × 10 ⁻⁵), indicating a greater degree of dissociation into H ⁺ ions at equilibrium (1)	2
39(b)	The methyl group (-CH ₃) in ethanoic acid is electron-donating (+I inductive effect) (1); This increases electron density on the carboxylate group, destabilising the CH ₃ COO ⁻ anion and strengthening the O-H bond in CH ₃ COOH (1)	2
40(a)	An internal salt / dipolar ion having separate positive and negative charges on different atoms within the same neutral molecule (1)	1
40(b)	Λ+H ₃ N-CH ₂ -COOH (or Cl ⁻ Λ+H ₃ N-CH ₂ -COOH) (1)	1
40(c)	H ₂ N-CH ₂ -COO ⁻ (or H ₂ N-CH ₂ -COO ⁻ Na ⁺) (1)	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
41(a)	Curve A (HCl): pH = 1.0 (1); Curve B (CH ₃ COOH): pH ~ 2.9 – 3.0 (1)	2
41(b)	Curve A: pH = 7.0 (1); Curve B: pH ~ 8.5 – 9.0 (1)	2
41(c)	At equivalence, sodium ethanoate, CH ₃ COONa, is formed (1); The ethanoate ion hydrolyses water: CH ₃ COO ⁻ + H ₂ O ⇌ CH ₃ COOH + OH ⁻ , producing excess OH ⁻ ions (1)	2
42(a)	Phenolphthalein (1); The equivalence point occurs at pH ~ 8.5–9.0, which falls squarely within the vertical pH range and transition range of phenolphthalein (8.3–10.0) (1)	2
42(b)	Methyl orange (1); The equivalence point of a weak base–strong acid titration occurs in the acidic region (pH ~ 5.0), matching the vertical drop and range of methyl orange (1)	2
43(a)	A weak acid (e.g. CH ₃ COOH) (1); and its conjugate base / salt with a strong base (e.g. CH ₃ COONa) (1)	2
43(b)	CH ₃ COO ⁻ (aq) + H ⁺ (aq) → CH ₃ COOH(aq) (1); The high reservoir of ethanoate ions combines with added H ⁺ to form undissociated ethanoic acid, maintaining [H ⁺] and pH almost constant (1)	2
44(a)	K _a = ([CH ₃ COO ⁻][H ⁺]) / [CH ₃ COOH] (1); Units: mol dm ⁻³ (1)	2
44(b)	Water is in enormous excess so its concentration (~55.5 mol dm ⁻³) remains effectively constant and is subsumed into the constant K _a (1)	1
45(a)	Na ₂ O: Basic (1); Al ₂ O ₃ : Amphoteric (1); SO ₃ : Acidic (1)	3
45(b)	Na ₂ O(s) + H ₂ O(l) → 2NaOH(aq) (1); pH ~ 13–14 (strongly alkaline) (1)	2
45(c)	Acid: Al ₂ O ₃ + 6HCl → 2AlCl ₃ + 3H ₂ O (1); Alkali: Al ₂ O ₃ + 2NaOH + 3H ₂ O → 2NaAl(OH) ₄ (or Al ₂ O ₃ + 2OH ⁻ → 2AlO ₂ ⁻ + H ₂ O) (1)	2
46(a)	HCN is an extremely weak acid that is almost completely undissociated in water (1); Significant energy is absorbed (endothermic process) to break H–CN bonds and dissociate HCN molecules during neutralisation (1)	2
46(b)	ΔH _{neut} (HCN) = ΔH _{dissoc} (HCN) + ΔH _{neut} (strong) ⇒ -11.7 = ΔH _{dissoc} + (-57.1) (1); ΔH _{dissoc} (HCN) = -11.7 - (-57.1) = +45.4 kJ mol ⁻¹ (1)	2
47(a)	Chlorine is an electronegative atom that exerts an electron-withdrawing inductive effect (-I effect) (1); This pulls negative charge away from the O–H bond, weakening it and facilitating proton loss (1); It also disperses the negative charge over the chloroethanoate anion (CH ₂ ClCOO ⁻), stabilising the conjugate base (1)	3
47(b)	Fluoroethanoic acid is stronger (1); Fluorine is more electronegative than chlorine (4.0 vs 3.0), creating a stronger electron-withdrawing inductive effect that stabilises the conjugate base even more (1)	2
48(a)	pH < 7 / acidic (1); NH ₄ ⁺ ion acts as a weak acid and hydrolyses water: NH ₄ ⁺ + H ₂ O ⇌ NH ₃ + H ₃ O ⁺ , producing excess hydronium ions (1)	2
48(b)	pH > 7 / alkaline (1); CH ₃ COO ⁻ ion acts as a weak base and hydrolyses water: CH ₃ COO ⁻ + H ₂ O ⇌ CH ₃ COOH + OH ⁻ , producing excess hydroxide ions (1)	2
49(a)	The methyl group (-CH ₃) is electron-donating (+I inductive effect) (1); This increases electron density on the nitrogen atom, making its lone pair more available to accept a proton (1)	2
49(b)	The nitrogen lone pair overlaps with and delocalises into the benzene ring's π electron cloud (1); The lone pair is less localised on nitrogen and therefore far less available to accept a proton (1)	2
50(a)	p(NH ₃) = 0.150 × 2.00 × 10 ⁴ = 3.00 × 10 ³ kPa (1)	1
50(b)	Remaining mole fraction = 1.00 - 0.150 = 0.850; x(N ₂) = 0.850 / 4 = 0.2125, x(H ₂) = 3 × 0.2125 = 0.6375 (1); p(N ₂) = 0.2125 × 2.00 × 10 ⁴ = 4.25 × 10 ³ kPa; p(H ₂) = 0.6375 × 2.00 × 10 ⁴ = 1.275 × 10 ⁴ kPa (1); K _p = (3.00 × 10 ³) ² / [4.25 × 10 ³ × (1.275 × 10 ⁴) ³] = 9.00 × 10 ⁻⁶ / [4.25 × 10 ³ × 2.073 × 10 ¹²] = 1.02 × 10 ⁻⁹ kPa ⁻² (1)	3
50(c)	NH ₃ (aq) + H ₂ O(l) ⇌ NH ₄ ⁺ (aq) + OH ⁻ (aq) (1); Conjugate acid: NH ₄ ⁺ ; Conjugate base: OH ⁻ (1)	2
50(d)	Thermodynamics favours low temperature (exothermic forward) and high pressure (fewer gas moles) (1); However, low temperature yields unacceptable rates; 450 °C with iron catalyst achieves acceptable rate while 200 atm is an economical pressure that avoids excessive plant construction costs (1)	2